Aditya Chaturvedi

Cham, Germany / +49 15226925116 / aditya.1540011@gmail.com / www.linkedin.com/in/aditya1540011

Profile

Master's student in Mechatronics and Cyber-Physical Systems with a strong interest in 3D computer vision, image processing, and intelligent systems. Experienced in working with 3D data and digital models, including CAD modeling and Unity-based virtual environments. I bring strong analytical and problem-solving skills, a structured work style, and high motivation to contribute to innovative computer vision and imaging technologies. Fluent in English with good German skills (B1).

Education

Deggendorf Institute of Technology — Deggendorf, Germany

M. Eng. Mechatronics and Cyber-Physical Systems — Grade-1.8

Technical Skills

XR & 3D Development: Unity (VR development, interactive simulation), Blender (3D modelling, asset optimization), VR XR application development, 3D CAD integration into virtual environments

Programming & Scripting: Python, MATLAB, Simulink, Linux (basic)

Robotics & Automation: ROS (Robot Operating System), Gazebo Simulation, URDF Modelling, Autonomous Navigation, Sensor Integration (Camera, LiDAR)

Mechanical Design & Simulation: SolidWorks, CATIA, ANSYS (FEA), System Modelling, Topology Optimisation, Additive Manufacturing (FDM)

Core Competencies: Mechatronic System Design, Robotics Integration, Process Optimisation, Problem Solving, Continuous Improvement, Robotics integration and simulation

Professional Experience

Production Supervisor — Jul 2023 – Feb 2024

Rajshree Engineering Pvt. Ltd., Pithampur, India

- Supervised manufacturing operations and ensured efficient production workflow and machine utilization.
- Performed technical defect analysis and implemented corrective actions to reduce downtime.
- Led continuous improvement initiatives using Kaizen and PDCA methodologies.
- Optimised production resources using the 4M methodology (Man, Machine, Material, Method).

Graduate Engineer Trainee — Mar 2022 – May 2023

TATA Motors, Pune, India

- Assisted in design validation and functional testing of automotive mechanical components to ensure compliance with engineering specifications.
- Implemented Lean Manufacturing principles to identify process bottlenecks and significantly reduce operational waste on the production line.
- Authored detailed technical documentation, including daily production reports, defect analysis logs, and project status updates to support cross-functional teams.

Graduate Engineer Trainee — Mar 2021 – Sep 2021

Wonder Cement, Rajasthan, India

- Acquired hands-on expertise in the cement manufacturing lifecycle through rigorous training and on-site operational involvement.
- Supported production planning strategies and monitored daily production operations to ensure workflow continuity and efficiency.
- Managed technical documentation and generated comprehensive operational reports to assist management in process tracking.

Academic Projects

VR Safety Training for 3D Printer Operations — Oct 2024 - Jan 2025

Deggendorf Institute of Technology, Cham

- Developed an immersive VR training application using Unity to simulate safe operation procedures for industrial Stratasys 3D printers.
- Created and optimized high-fidelity 3D assets using Blender and SolidWorks, ensuring real-time performance in a VR environment.
- Implemented interactive training scenarios and hazard simulations to enhance operator learning and safety awareness.
- Tools: SolidWorks, Blender, Unity.

Dryer Door Push Latch Mechanism Oct 2024 - Jan 2025

Deggendorf Institute of Technology, Cham

- Designed and topology-optimised a push-lock/push-release dryer latch mechanism using SolidWorks, engineered to withstand high operating temperatures and a lifecycle of 28,000 cycles.
- Prototyped the component via Additive Manufacturing (FDM) using ABS material, optimising the design for weight reduction, ease of assembly, and cost-efficient manufacturing.
- Tools: SolidWorks, Additive Manufacturing (FDM), Topology Optimisation, Ultimaker Cura

Case Study Cooperative and Autonomous Systems Mar 2024 – Jul 2024

Deggendorf Institute of Technology, Cham

- Designed and modeled a differential-drive robot in SolidWorks and converted it into a URDF simulation model.
- Implemented robot simulation and navigation algorithms using ROS and Gazebo.
- Integrated LiDAR and camera sensors for environmental perception and obstacle avoidance.
- Conducted simulation experiments to evaluate robot mobility and autonomous navigation performance
- Tools: SolidWorks, ROS, Gazebo, URDF

Case study Mechatronic System Simulation Mar 2024 – Jul 2024

Deggendorf Institute of Technology, Cham

- Simulation of the soil's moisture content by comparing bare soil and mulched soil.
- Approx. 20% more moisture retention in mulched soil in comparison to bare soil.
- SolidWorks, Gazebo, ROS

Design Analysis and Fabrication of a Single Piston Floating Calliper of the Hydraulic Brake System of an All-Terrain Vehicle 2018 –2019

KIET Group of Institutes, India

- To get the required amount of brake force for our customised vehicle.
- Modular Al 7075-T6 calliper design reduced thermal deformation, machining cost, and thermal displacement by 8.56% vs. conventional.
- SolidWorks, Ansys

Achievements & Honors

- Secured All India Rank 07 and 1st position in CAE and Sled Pull at SAEINDIA BAJA 2018.
- Secured All India Rank 24 and 3rd position in the Cost Event at SAEINDIA BAJA 2017.
- Active SAEINDIA member (2015–2018) with participation in national-level engineering competitions and technical events.
- 3rd Place – CAD Competition (Autodesk Inventor)

Interests

- Focus Areas: Robotics and autonomous systems, mechatronic system design, intelligent automation, simulation and modeling, additive manufacturing
- Continuous Learning: Active learner in robotics, cybersecurity fundamentals, and automation technologies through academic projects, case studies, and self-directed technical exploration
- Multilingual: Professional English proficiency, German B1 (intermediate), experienced in cross-cultural academic and industrial collaboration.
- XR Technologies (AR/VR), 3D simulation and digital twins, Industrial VR training systems, Metaverse applications in manufacturing, Robotics simulation